

Online Retail Sales Performance Analysis Report

1. Project Overview

This project analyzes an online retail dataset using Microsoft Excel and Power BI to understand sales performance, customer behaviour, product performance and country-wise sales trends. The dataset was cleaned in Excel before being imported into Power BI, where interactive dashboards were developed using DAX measures and visualizations.

2. Tools Used

- Microsoft Excel
- Power BI Desktop
- DAX (Data Analysis Expressions)

3. Data Cleaning Process

- **Removed Duplicate Records**
Removed 5,268 duplicate records using Excel's Remove Duplicates feature, leaving 536,641 records.
- **Removed Cancelled Orders**
Removed 9,251 cancelled transactions identified by InvoiceNo beginning with "C".
- **Removed Negative Quantity Records**
Removed 1,336 records with negative quantities after validating remaining transactions.
- **Removed Zero-Priced Transactions**
Removed 1,176 records where UnitPrice was zero because they did not contribute to sales revenue.
- **Handled Missing Values**
Verified there were no missing product descriptions. Records with missing CustomerID values were retained because they represented valid sales transactions.

4. Dashboard Development

Two interactive dashboards were developed using Power BI.

Dashboard 1 – Executive Sales Dashboard

The Executive Dashboard provides a high-level overview of sales performance.

It includes:

- Total Sales
- Total Orders
- Total Quantity Sold
- Total Customers
- Monthly Sales Trend
- Sales by Country
- Top 10 Products
- Interactive Filters (Country, Year, Month)

Dashboard 2 – Customer Analysis Dashboard

The Customer Dashboard focuses on customer purchasing behavior.

It includes:

- Total Customers
- Average Revenue per Customer
- Average Order Value
- Top 10 Customers by Sales
- Top 10 Customers by Orders

5. Dashboard Features

- Executive Sales Dashboard with KPI cards for Total Sales, Total Orders, Total Quantity and Total Customers.
- Monthly Sales Trend analysis.
- Sales by Country visualization.
- Top 10 Products by Sales.
- Customer Analysis dashboard with Average Revenue per Customer, Average Order Value, Top 10 Customers by Sales and Top 10 Customers by Orders.
- Interactive slicers for Country, Year and Month.

6. Business Insights

- The United Kingdom generated the highest share of total sales, making it the company's primary market.
- Sales increased steadily from September to November, with November recording the highest sales.
- A small number of products contributed a significant portion of total revenue.
- A limited number of customers generated a large share of overall sales.
- The average revenue per customer was approximately 2.45K, while the average order value was approximately 533.

7. Business Recommendations

1. Increase inventory before the September–November peak sales season.
2. Strengthen customer retention through loyalty programmes for high-value customers.
3. Expand marketing activities outside the United Kingdom to diversify revenue sources.
4. Focus on promoting top-selling products while improving lower-performing products.
5. Use seasonal sales trends for forecasting and inventory planning.

8. Conclusion

The project demonstrates the complete process of cleaning, analysing and visualising retail sales data using Excel and Power BI. The dashboards provide meaningful insights into sales trends, customer behaviour and product performance, supporting better business decision-making.

Dashboard Visuals

Online Retail Sales Dashboard

Country

All

Year

All

Month

All

Total Sales

11M

Total Customers

4K

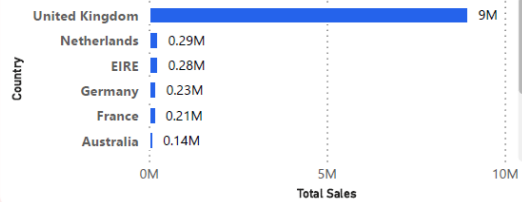
Total Quantity

6M

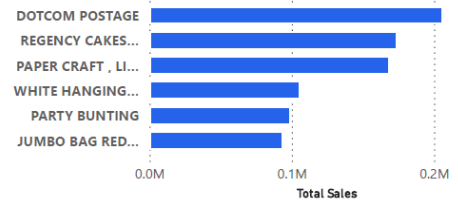
Total Orders

20K

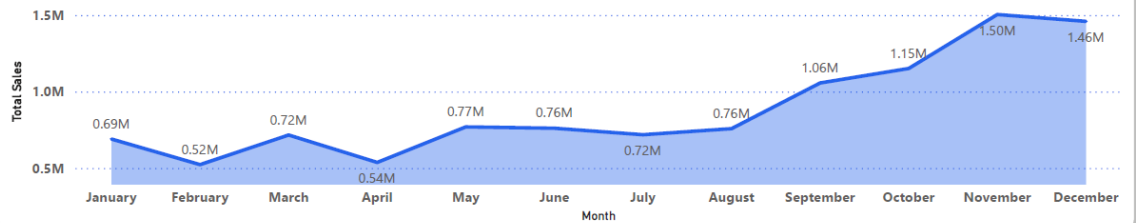
Sales by Country



Top 10 Prodcuts



Monthly Sales Trend



Prepared by: Anzar Ali Warsi | Future Interns Data Science & Analytics Task 1 | 2026

Customer Analysis

Average Revenue Per Customer

2.45K

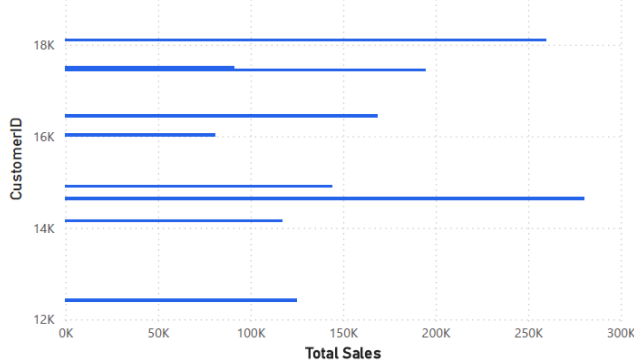
Total Customers

4K

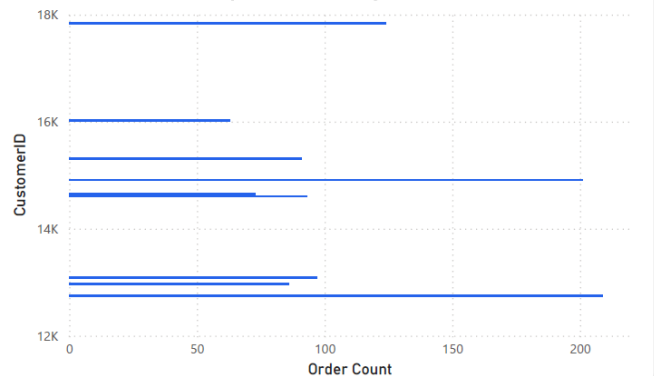
Average Order Value

533

Top 10 Customers by Revenue



Top 10 Customers by Orders



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